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Rotary Separation Apparatus and Process of Use

Abstract

A rotary separation device deploys a drum with mesh like opening on the cylindrical surfaces and a removable cover or cap for filling in an upright position and removal of product or spent matter in an inverted position. When the drum is loaded with material, and the cover closed, it is rotatable to a horizontal position, and disposed in an outer container. The drum is rotated in the horizontal position to initiate the separation process. The outer container may be formed by the mating engagement at a common rim of an upper and lower vessel that form the sealed container.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation application of U.S. patent application Ser. No. 17/782,662, filed on Jun. 5, 2022, which is a national stage entry of International/PCT application Serial No. PCT/US20/63631, filed on Dec. 7, 2020, which claims the benefit of priority to the U.S. Provisional Patent Application that was filed on Dec. 5, 2019, having application No. 62/943,925, each of which is incorporated herein by reference.

BACKGROUND OF INVENTION

[0002] The field of inventions is processing plant and botanical materials to extract components and separating the components by size, and more particularly a rotary separation apparatus that contains the plant or botanic material.

[0003] Rotary separation machine deploy an outer chamber or container that forms an outer bound space for collecting material that exits through screens, perforation or mesh that forms an outer portion of an inner bound chamber that is rotated to agitate and tumbled the plant or botanical materials. The tumbling, agitation and shear that may be provided to the screen, mesh perforation and/or other contents and conditions may further breakdown the plant matter to isolate more desired components that can then be separated by size from desired plant matter. The rotation process will continuously expose the contents of the inner bound container to the perforated covering so matter smaller than the holes or perforation can pass through to be collected in the outer chamber. Depending on the plant or botanical matter and the intended purpose, it may be desirable to utilize the material that is in the inner bound chamber or that which exits it, as well as both these materials, which can then undergo different process or uses.

[0004] The inventor has discovered various improvements in equipment and processes to process plant and botanical matter that are disclosed in the commonly owned U.S. Pat. No. 10,512,938B2 which issued Dec. 24, 2019, and U.S. Pat. No. 10,507,223B2 which issued Dec. 17, 2019, which are incorporated herein by reference.

[0005] The productivity, capacity and cost effectiveness of these processes can be improved as disclosed in the apparatus and methods disclosed below.

[0006] The above and other objects, effects, features, and advantages of the present invention will become more apparent from the following description of the embodiments thereof taken in conjunction with the accompanying drawings.

SUMMARY OF INVENTION

[0007] In the present invention, the first object is achieved by providing a rotary separation apparatus that comprises a drum having a generally cylindrical shape with a primary axis coincident with the principal axis of the cylinder, in which the drum has lateral surface with a plurality of perforations and a base of the cylinder that is opposed to an upper of the cylinder, a container that forms an enclosed space around the drum when the drum is disposed with the principal axis in a horizontal orientation, one of more actuators that are operative modulate the orientation of the principal axis of the cylinder from between an upright and inverted vertical orientation and dispose the drum horizontally in the container, a means to rotate the drum about the principal axis when the drum is disposed horizontally within the container.

[0008] A second aspect of the invention is characterized by such a rotary separation apparatus in which the drum has a removable cap that closes an opening in the upper end of the cylinder.

[0009] Another aspect of the invention is characterized by any such rotary separation apparatus in which the container forms the enclosed space when an upper and lower vessel are brought into mutual engagement at a rim of the upper vessel to a rim of the lower vessel.

[0010] Another aspect of the invention is characterized by any such rotary separation apparatus in which the lower vessel is capable of being inverted from an upright horizontal orientation to a downward facing horizontal orientation by the one or more actuators to empty the contents thereof via the opening in the upper end of the cylinder.

[0011] Another aspect of the invention is characterized by any such rotary separation apparatus when the one of more actuators are operative to modulate the orientation of drum from being disposed with the principal axis of the cylinder horizontally within the container to vertically outside the container by rotating the upper vessel away from the lower vessel to dispose both the upper vessel and the lower vessel in an upward facing orientation and then rotating the drum to dispose the principal axis of the cylinder vertically.

[0012] Another aspect of the invention is characterized by any such rotary separation apparatus wherein the one of more actuators includes a first motor and second motor in which the first motor is operative to rotate the upper vessel away from the lower vessel and the drum is rotated between the vertical and horizontal orientation of the principal axis of the cylinder by the second motor the coupled to a support frame that is external to common housing that contains the first motor to rotate the common housing at least 60 degrees.

[0013] Another aspect of the invention is characterized by any such rotary separation apparatus wherein the means to rotate the drum about the principal axis when the drum is disposed horizontally within the container is a primary motor coupled to the common housing to rotate a shaft that extends through the common housing to connect to the base of the cylinder in which the shaft extends in the direction of the principal axis of the cylinder.

[0014] Another aspect of the invention is characterized by any such rotary separation apparatus wherein the one of more actuators includes a third motor that is connected to the support frame and the second motor is mounted at a distal end of a primary arm and the proximal end of the primary arm is connected to the third motor in which the third motor is operative to rotate the primary arm from a horizontal to a vertical orientation before the one of more actuators are operative modulate the orientation of the principal axis of the cylinder from an upright to an inverted vertical orientation

[0015] Another aspect of the invention is characterized by any such rotary separation apparatus wherein the container is sealed by a portal with a cover that supports an axle that is coupled to rotate the drum about the principle axis and the apparatus is configured to remove the drum via the portal and disposes a motor on the opposite side of the cover.

[0016] Another aspect of the invention is characterized by any such rotary separation apparatus in which the container comprises in a lower portion a drawer disposed to collect matter that exits the drum via the plurality of perforation in which the drawer is configured translate away from the lower portion to provide access to the collected matter.

[0017] Another aspect of the invention is characterized by any such rotary separation apparatus wherein the one or more actuators is operative to remove the container laterally via a side portal such that it translates in the direction of the principal axis.

[0018] Another aspect of the invention is characterized by any such rotary separation apparatus wherein the upper and lower vessel are configured to be separated to remove the drum by a rotary joint that is spaced apart from the rim of both the upper and lower vessel such that the inverted upper vessel is spaced apart from the lower vessel when the drum is removed from the opened container.

[0019] Another aspect of the invention is characterized by a system for processing materials in a rotary separation apparatus, in which the rotary separation apparatus comprises a drum having a generally cylindrical shape with a primary axis coincident with the principal axis of the cylinder, in

which the drum has lateral surface with a plurality of perforations, a container that forms an enclosed space around the drum when the drum is disposed with the principal axis in a horizontal orientation, one of more actuators to invert the drum when out of the container and dispose the drum horizontally in the container, a means to rotate the drum about the principal axis when the drum is disposed horizontally within the container, at least a first conveyor belt configured to transport material to an opening in an upper end of the cylinder when the drum is oriented with the principal axis in the vertical orientation.

[0020] Another aspect of the invention is characterized by such a system for processing materials in a rotary separation apparatus in which the container that forms the enclosed space around the drum comprises an upper and lower vessel in which the upper vessel is disposed over and in sealed engagement to the lower vessel when the drum is disposed with the principal axis of the cylinder horizontal when within the container and further comprising a second conveyor belt disposed to receive material from the lower vessel when upper vessel is removed from the lower vessel and lower vessel is inverted from an upright horizontal orientation to a downward facing horizontal orientation.

[0021] Another aspect of the invention is characterized by any such system for processing materials in a rotary separation apparatus wherein the first conveyor belt is operative to lift material from a hopper to at least the height of the open end of the container such that material on exiting an end of the conveyor distal from the hopper falls into the container.

[0022] Another aspect of the invention is characterized by a process for treating materials, the process comprising the steps of providing a drum having perforation on lateral surfaces and an upper surface with a closable opening, the drum being defined by a cylindrical axis that is disposed orthogonal to the upper surface and a base that is disposed opposite the upper surface and is also disposed parallel to the lateral surfaces, providing a container to surround the drum when disposed in an orientation when the cylindrical axis is closer to a horizontal orientation than a vertical orientation, charging the drum with material to be processed via the opening when the drum is outside the container and the opening is disposed above the base, closing the opening, disposing the drum with the cylindrical axis is closer to a horizontal orientation than a vertical orientation, rotating the drum within the container, removing the drum from the container, orienting the drum with the base disposed above the opening to discharge material from the container.

[0023] Another aspect of the invention is characterized by such a process for treating materials further comprising a step of removing material that traversed the drum lateral surfaces via the perforations from the container.

[0024] Another aspect of the invention is characterized by any such process for treating materials where the material that traversed the drum lateral surfaces via the perforations is removed from the container by one of a drawer or shoot that translates or pivot away from a lower portion of the container.

[0025] Another aspect of the invention is characterized by any such process for treating materials where the material that traversed the drum lateral surfaces via the perforations is removed from the container by opening the container along a seam that separates connected portions and tilting at least on connected portion by rotation to pour out the material.

[0026] Another aspect of the invention is characterized by any such process for treating materials wherein the container is separated at a lateral seam to dispose a lower vessel under the drum that collects the material separated by the drum.

[0027] Another aspect of the invention is characterized by any such process for treating materials wherein the step of removing the drum from the container occurs after the step of orienting the drum with the base disposed above the opening to discharge material from the container.

[0028] In other aspects of the invention, the upper and lower vessels are configurable to use different separation processes and media. A resulting product may be retained in the drum by the mesh or pass through the mesh to enter the lower vessel. This product can be removed from a

vessel or drum by inverted it after the vessels are separated from the drum. In some embodiments, the drum, upper and lower vessel are manipulated by a common multi-axis actuator that is capable of both separating the drum from each vessel and the selective orientation of each component as necessary for the removal of the resulting product, as well as then refilling the drum to repeat the separation process. The common multi-axis actuator may also rotate the drum in the horizontal position. The separation process may be semi-continuous via the rapid repetition of batch operations in which the spacing and orientation of the vessels and the drum before and after the separation process permit feeding of material, product and spent matter by different conveyor belts. [0029] The above and other objects, effects, features, and advantages of the present invention will become more apparent from the following description of the embodiments thereof taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0030] FIG. 1A to 1E is a schematic diagram to illustrate a sequence of steps in an embodiment of the apparatus and method, in which FIG. 1A-D are schematic side elevations and FIG. 1E is a top plan view.

[0031] FIG. 2A to 2D is a schematic diagram to illustrate a related sequence of steps in an embodiment of the apparatus and method, in which FIG. 1A-D are schematic side elevations

[0032] FIG. 3 is a front perspective view of another embodiment of the invention.

[0033] FIG. 4 is front elevation view of the embodiment of FIG. 3 in which the container is closed around the drum.

[0034] FIG. 5 is a perspective view of the embodiment of FIG. 3 in which the drum is disposed for loading as in FIG. 1D and FIG. 2D.

[0035] FIG. 6 is a front elevation view of the apparatus in the configuration in FIG. 5.

[0036] FIG. 7 is a side elevation view of the apparatus in the configuration in FIG. 5.

[0037] FIG. 8 is a rear cross-sectional elevation view of view of the apparatus in the configuration in FIG. 5 taken at the broken section line in FIG. 6.

[0038] FIG. 9 is an expanded perspective view of an embodiment of an actuator.

[0039] FIG. 10 is a partially exploded perspective view of the actuator of FIG. 9.

[0040] FIG. 11 is a front perspective view of another embodiment of the invention that includes a pair of conveyor belts disposed to receive product from each of the vessels that form the container.

[0041] FIG. 12 is a front perspective view of another embodiment of the invention that includes a different pair of conveyor belts disposed to receive product from each of the vessels that form the container and a conveyor to fill the drum.

[0042] FIG. 13 is a schematic perspective view of another embodiment of the apparatus.

[0043] FIG. 14 is a schematic side elevation view of the embodiment of the apparatus in FIG. 13 illustrating alternative positions and orientations of the drum in and outside of the container.

[0044] FIG. 15A is schematic perspective view of another embodiment of the apparatus illustrating moving between alternative positions inside and outside. FIG. 15B is a schematic sectional view of the drum in the container whereas in FIG. 15C the drum is schematically illustrated in a section view outside the container and rotated to the vertical orientation.

[0045] FIG. 16 is a flow chart of an embodiment of the method illustrated in FIG. 1A-E and 2A-D.

DETAILED DESCRIPTION

[0046] Referring to FIGS. 1 through 16, wherein like reference numerals refer to like components in the various views, there is illustrated therein a new and improved Rotary Separation Apparatus, generally denominated **1000** herein.

[0047] In accordance with the present invention the Rotary Separation Apparatus **1000** comprises a

drum **100** having a generally cylindrical shape with a primary axis **100a** coincident with the principal axis of the cylinder, in which the drum **100** has lateral surface **110** with a plurality of perforations **111**. A container **200** forms an enclosed space around the drum **100** when the drum **100** is disposed with the principal axis **100a** in a generally horizontal orientation. One or more actuators **300** may invert the drum **100** when out of the container **200** to dispose the drum **100** generally vertically and dispose the drum **100** horizontally within the container **200**. The rotary separation apparatus **1000** also comprises a means to rotate the drum **100** when the drum **100** is disposed horizontally within the container **200**.

[0048] It should be understood that the terms horizontal and vertical are not limited to strictly parallel and perpendicular to the ground but may embrace variation of at least 5-30 degrees and still provide the intended functions and resulting benefits of the process for treating materials **1600**. Process **1600** may comprise the steps of providing a drum **100** having perforation **111** on lateral surfaces and an upper surface with a closable opening, the drum being defined by a cylindrical axis that is disposed orthogonal to the upper surface and a base that is disposed opposite the upper surface and is also disposed parallel to the lateral surfaces (step **1605**), providing a container **200** to surround the drum **100** when disposed in an orientation when the cylindrical axis is closer to a horizontal orientation than a vertical orientation (step **1610**), charging the drum **100** with material to be processed via the opening when the drum is outside the container and the opening is disposed above the base (step **1615**), closing the opening (step **1620**), disposing the drum **100** with the cylindrical axis **100a** is closer to a horizontal orientation than a vertical orientation (step **1625**), rotating the drum **100** within the container **200** (step **1630**), removing the drum **100** from the container **200**, orienting the drum **100** with the base disposed above the opening to discharge material from the drum (step **1635**).

[0049] The perforation **111** may be provided by forming the lateral surface from **110** from a plurality of panel that are assembled into a frame work, each panel being formed of rigid sides that form a supporting frame, with a screen, mesh or perforated plate(s) extending across the frame. Alternatively, the perforation **110** may be provided by wrapping a flexible screen or mesh around the entire lateral surface **110**, or across each panel. The lateral surface **110** can be continuously curved, curvilinear or a faceted, such as a multisided polygon, and combinations thereof and the like.

[0050] In preferred embodiments, the container **200** is formed of an upper **210** and a lower **220** vessel that mate at a common junction of their rims **211** and **221**, respectively. A gasket **215** may be disposed on one of the rims **211** or **221** such that a fluid tight seal is formed when the one or more actuator(s) **300** is deployed to urge the rims **221** and **211** into contact.

[0051] The vessels **210** and **220** may have any shape and are preferably shapes to contain a fluid when the rim **118** is disposed above the interior of the container **200** and the upper **210** and lower vessel **220**. However, to reduce the size and facilitate separating the vessels **210** and **220**, it is preferred that are shaped generally like the drum **100** but larger, leaving a space to provide media, such as a freezing agent, that aids in the separation process, as well as to collect the components or product that exits through the perforations **111**. It may be desirable to provide a deeper lower vessel **220** to provide more capacity for separated matter.

[0052] In some processes, the desired product is retained in the drum **100**, while a less valuable or waste product passes through the perforations **111** when the drum **100** is rotated about the primary axis **100a**. In other processes, the desired product exits the perforation in the drum **100**.

[0053] In some embodiments, there is a means to remove fluid and gas from the container **200**. For example, in FIG. 4-6, lower vessel **210** has an outlet **219** to allow expanding gas from a fluid or solid freezing agent to escape the closed container **200**, such as via a hose (not shown) to be discharged safely away from personnel.

[0054] While the apparatus is not intended to be limited to any particular size drum **100**, it is most advantageous for drums of more than about 5 gallons capacity, which become awkward and

difficult to manually manipulate. More preferably, the drums have capacities of 50 or more gallons, and optionally hundreds of gallons. By drum capacity, we mean the volume of fluid that could be contained therein if the lateral surfaces **110** were not perforated. Moreover, the term drum is not intended to be limited to any aspect ratio of range aspect ratios of the diameter of the base to the height in the direction of the cylindrical axis.

[0055] It should be appreciated that the apparatus **1000** is configured to rapidly process repeated large batches of matter in the drum **100** to provide a high throughput and enhanced economy of scale. It has been appreciated by the inventor that as the drum is enlarged to process larger batches of material, a limit to overall productivity is delays in loading and unloading the drum **100**, as well as the end products, which are either retained in the drum **100** or exit the perforation **111** in the drum **100** to be retained in the container **200**.

[0056] Accordingly, various embodiments provide for the apparatus **1000** to first orient the drum **100** with the opening of side **100e** at the rim **118** disposed above the base **100b** of the drum **110** to gain the advantage of gravity to feed starting material into the drum **100**. The apparatus **1000** is then operative to dispose the drum **100** with the cylindrical axis **100a** in a generally horizontal orientation for rotation by the shaft **116** connected to the base **100b** to effect separation. Thereafter, when the process of separation is complete the drum **100** can be inverted from the loading vertical orientation, so spent material falls out of the drum **100** opening defined by the rim **118** thereof at end **100e**.

[0057] It should be noted that when the drum **100** is rotated in a generally horizontal orientation product that exits the perforations **111** will fall downward into the lower vessel **220**.

[0058] Hence, to further improve process efficiency in some embodiment it is also

[0059] desirable to invert the lower vessel **220** to use gravity to remove the separated product, but in most embodiments after the drum **100** is removed from the lower vessel **220** vessel that forms a part of the container **200**.

[0060] Further, the apparatus **1000** and methods of use described herein may use several drums **100** in association with a single container **200**, as one drum **100** may be filled or emptied of matter with the alternate identical drum is being rotated with the container **200**.

[0061] It is also preferable that at least the lower vessel **220** has an expanse **228** below the rim **221** with a funnel like shape for aid in the flow of solid or fluid material to converge at a sealable drainage portal **229**. The drainage portal **229** can thus remove solids, slurries of solid in a fluid, and fluid from the container **200** without inversion. This drainage portal **229** preferably has an external coupling and valve, such as shown at outlet **219**. FIG. 5 illustrates a non-limiting embodiment of the expanse **228** of the lower vessel **220** with an irregular polygonal shape below the rim **221**.

Outlet **219** is at the lowest elevation of the multiple downward sloping polygonal slides that form the expanse **228**. The sides that form the expanse **228** may be any combination of linear, curvilinear, and curved shapes to aid in the flow of solids and liquids to the drainage portal **229**.

[0062] In the embodiments of FIG. 1A-E, FIG. 2A the container **200** is closed and surrounds the drum **100** when the drum is disposed is spun about axis **100a** in the separation process. Preferably container **100** has a means to admit a freezing agent before or during this process. The freezing agent is preferably inert and can be admitted as a gas, fluid or solid into the cavity between the closed container **200** and the drum **100**, as well as directly into the drum **100**. Various means of admitting such freezing agents into a rotary separation apparatus are disclosed in the commonly owned U.S. Pat. No. 10,512,938B2 which issued Dec. 24, 2019, and which is incorporated herein by reference. Further, commonly owned U.S. Pat. No. 10,507,223B2 which issued Dec. 17, 2019, also discloses the process conditions for using a rotary separation apparatus for various application and is also incorporated herein by reference.

[0063] After the separation process is complete, it is necessary to remove product from either or both the drum **100** and container **200**, and then recharge the drum **100** with fresh material to be processed, as illustrated in FIGS. 1D and 2D.

[0064] FIGS. 1B and 1E illustrate the upper vessel **220** being rotated at a side hinge **301** to invert and be spaced adjacent the lower vessel **201**. The hinges **301** and **302** are spaced away from the vessel **210** and **220** by arms **303/304**. The ends of the arms **303/304** may have motors or gears **305/306** to rotate and invert either or both vessels **210** and **220**.

[0065] It should be noted in FIG. 2B that a means to engage and separate the upper and lower vessels may deploy an edge hinge **302** that connects the common rims at a side parallel with the primary axis **100a**. Alternatively, as in the embodiments of FIG. 1A-D and FIG. 3-12, the vessels **210** and **220** may separate at a hinge **301** disposed at the common edge of the rim transverse to the primary axis **100a**. In any such means, the hinges **301** and **302** comprise a rotary joint that is preferably spaced apart from the rim of both the upper and lower vessel such that the inverted upper vessel is spaced apart from the lower vessel when said means has been deployed.

[0066] In step 1C or 2C, the drum **100** can also be lifted, moved, and inverted by well known machines used to handle and manipulate industrial and storage drums that are solid to contain fluid contents, as well as other material. Non limiting examples of such machines are disclosed in the following US Patent documents that are incorporated herein by reference: U.S. Pat. No. 2,645,372 (issued to Broesdma, J. C. on 1953 Jul. 14) U.S. Pat. No. 3,410,431 (issued to Vik, A. M. in 1968 Nov. 12), U.S. Pat. No. 3,448,800 (issued to Howard, W. E. on 1969 Jun. 10), U.S. Pat. No. 6,0245,529 (issued to Kristensen, A. on 2000 Feb. 15) and U.S. Pat. No. 10,450,176 (issued to Beard, T. E. on 2019 Oct. 22). The machines disclosed in these published patents or applications show various ways to provide a capability to lift, move and invert drums or barrels, as well as separating them from the opened container. These machines and the instant invention may use hydraulic actuators or motors, that is electric dynamo machines in places of each other.

[0067] It is also preferable to provide a means to invert at least one of the upper **220** and lower **210** vessel after they have been separated. It should be appreciated that the terms upper and lower are relative, as the vessels that form the container **200** can be inverted when the drum **100** is disposed therein.

[0068] It is also preferable to provide a means to invert both the upper **220** and lower **210** vessel, as shown in FIGS. 1D and 2D. This may be desirable to clean the vessels **210** and **220** from any interior residue, as well as for to clean with fluids and drain the fluid used in cleaning. The upper vessel **220** may also have a port **2291** that can be used for drainage or to locate a thermal sensor for monitoring the temperature of the contents as cooling by the freezing agent used in the process.

[0069] It is also preferable to provide a means to invert the drum **100** to discharge material from one end **100e** that configured to be opened at rim **118**. The end **100e** is disposed transverse to the principle axis **110a**, opposite base **100b**.

[0070] The drum **100** can then be reloaded with fresh matter for separation in this upright orientation. In the next step in the process, when the drum **100** has been reloaded with fresh plant or botanical matter and the opening closed, it is also preferable to provide a means to dispose the drum **100** in the lower vessel **220** and dispose the upper vessel **210** to engage the lower vessel **220** at the common rims drum **100** with a rotary drive means. Such rotary drive mean may include a shaft that passes through container **200** and engages a base of the drum.

[0071] In the embodiment of FIG. 3-12 various actuators provides the means to rotate the drum **100**, invert the drum **100** and dispose the drum **100** in the container **200**.

[0072] It should also be appreciated that in the embodiment of FIG. 3-12, a series of actuators have the means to rotate and invert the upper and lower vessel as well as the drum. In some embodiments these means include a common actuator **300**, which is illustrated in more detail in FIGS. 9 and 10, showing a common housing **305** that supports a cluster of motor that are configured for independent opening of the container **200**, removal of the drum **100**, and rotation of the vessel(s) **210/220** that form the container **220** to remove the contents, such as onto the conveyor **601** and **602** of FIG. 11-12. The drum **100** may be filled by another conveyer **603** by lifting material in hopper **604** to drop into the opening at the rim **118** when the cap or cover **117** is

removed or raised.

[0073] The common actuator **300**, which acts as hinge **301**, provides cantilevered support for each vessel **210** and **220** via pair of arms **310** and **320** that are each in hinged connection on opposing sides of the housing **305**. The arms **311** and **312** of pair **310** respectively connect opposing sides of the housing **305** to dual shafts of motor **313** fixed within the housing **305**. The housing **305** also support bearings for the motor shafts. The pairs of arms **310** and **320** are also in hinged connection to the housing **305** of the common actuator **300** and spaced apart from a central region thereof, being generally paced apart from the other arms **311** and **312** by the same distance from the central region. Either set of arms **311** and **312**, and or **321** and **322**, can rotated the connected vessel upward and away from the other vessel and/or the drum **100**, so the drum **100** can be accessed.

[0074] The opposing side of the arms **311** and **312** connect to a common plate **314** that supports another motor **315** that has a shaft coupled to the side of the lower vessel **210**. The motor **315** may be energized by a controller **530** to invert the lower vessel **210**. The controller **530** may energize all or some of the motors of the various actuator means in the various embodiment in the sequences illustrated in FIGS. 1A-E and 2A-D, among others.

[0075] The arms **321** and **322** respectively connect opposing sides of the housing **305** to dual shafts of motor **323**. The opposing side of the arms **321** and **322** from the housing **305** are connected to a common plate **324** that support another motor **325**. Motor **325** has a shaft coupled to the side of the upper container **220**. The motor **325** may be energized by the controller **530** to invert the upper container **220**.

[0076] A motor **115** at the bottom of the housing **305** has a shaft **116** that is coupled to the base of the drum **100**. The shaft **116** passes through the housing **305** between motor **313** and motor **323**. The motor **115** may be energized by the controller **530** to spin the drum **100** when it is enclosed in the container **200**, in which the upper **220** and lower **210** vessel engage at the gasket **215**.

[0077] An external motor **415** is provided to turn a shaft **416** that is coupled to the exterior of the housing **305** and is thus capable of rotating the drum **100** by 90 degrees from a vertical orientation of the axis **110a** in FIG. 6 to the horizontal orientation in other Figures. The external motor **415** is mounted at an end of a primary arm **510**

[0078] A motor **515** is disposed opposite end of a primary arm **510** than motor **415**. The axis of rotation of motors **415** and **515** are parallel. A hinge couple or bearing structure, which is optionally part of the motor **514** that is coupled to a support structure **500** such that the load of drum **100** and container **200**, along with actuator **300** are transferred to the ground. The primary arm **510** is configured to rotate raise or lower the drum **100** and container **200** when the motor **515** is energized by the controller **530**. The controller **530** may be mounted in the support structure or frame **500**.

[0079] The opposing end of arm **510** supports the external motor **415** that rotates the shaft **416** and can also rotate the common actuator **300** by 180 degrees relative to the position in which it is disposed the filling orientation with the rim **118** above the opposing side as in FIG. 3 among other. The primary arm **510** can rotate 90 degrees to raise the common actuator **300** and the attached drum **100** and container vessels **210** and **220**. Then the attached drum **100** can be inverted.

[0080] Hence, motor **515** together with motor **415** provide a means to invert the drum **100** after it is removed from the lower vessel **220** by raising the common actuator via arm **510**, being rotated to the vertical orientation by motor **515**. Then motor **415** can then rotate the shaft **416** by 180 degrees, thus inverting the housing **305**. The common actuator **300** may separate the upper **220** and lower **210** vessels by rotating one by at least about **120** to about 160 degrees with respect to the other.

[0081] In general, wherein the one of more actuators includes a first motor and second motor in which the first motor is operative to rotate the upper vessel away from the lower vessel and the drum is rotated between the vertical and horizontal orientation of the principal axis of the cylinder by the second motor the coupled to a support frame that is external to common housing that contains the first motor to rotate the common housing at least 60degrees.

[0082] The drum **100** has a cap or cover **117** over a portion of one end of what may be considered an upper base. The cap or cover **117** may be removed manually or by the controller **530** via energizing or de-energizing a latch hinge to open the drum **100** at rim **118**. The lower base of the drum **100** is couple to a rotary shaft **116** of the actuator **300** that is used to spin or rotate the drum at the desired speed within the container **100**.

[0083] FIG. **12** illustrated how the drum **100** may be supported upright with the cap or cover **117** released to provide the opening for filling via the conveyor **603**. While axis **100a** is vertical in this figure, the drum may also be filled when the axis **110a** is tilted upright 45 degrees from the horizontal positions, as well as intermediate positions. All of the collection baskets and source feeds, such as conveyors, may be supported on digital scale to provide for repeated filing of the drum **100** in each batch, as well as weighting the product of each production cycle to log the yield and other process variables.

[0084] The drum **100** is preferably instrumented with one or more thermal sensors to provide a consistent degree of cooling before the separation process starts.

[0085] FIGS. **13** to **15B** schematically illustrate variants of the apparatus **100** with alternative means to lift, move and turn/invert the drum **100**. In these embodiments the container **200** forms an enclosed space around the drum **100** when the drum **100** is disposed with the principal axis **100a** in a horizontal orientation

[0086] In FIG. **13-14**, the container **200** is formed of an upper **210** and a lower **220** vessel that mate via a hinge **301** at common junction of their respective rims **211** and **221**. A gasket **215** is disposed on one of the rims **211** or **221** such that a fluid tight seal is formed when the common actuator **300** is deployed to urge the rims **221** and **211** into contact.

[0087] The container **200** opens as in the embodiments of FIG. **1A-D** and FIG. **3-12**, such that the vessels **210** and **220** may separate at a hinge **301** disposed at the common edge of the rim transverse to the primary axis **100a**. The hinges **301** and **302** comprise a rotary joint that is preferably spaced apart from the rim of both the upper **210** and lower vessel **220** such that the inverted upper vessel **210** is spaced apart from the lower vessel **220** when said means has been deployed.

[0088] When the upper vessel **210** is inverted and removed from the lower vessel **220**, an actuator **1301** is configured to remove the drum from the container in the opposite direction, deploying a hinge mechanism **1302** to flip the drum **100** orientation 180 degrees, first to a vertical orientation and then horizontally, and optional to flip the drum **100** another 90 degrees to invert it.

[0089] After the drum **100** is filled via the opening formed when the cap **117** is removed from rim **118**, the actuator **1301** is operative to reverse the movement and return the drum **100** to the container **200**. The hinge mechanism **1302** coupled the drum base **110b** to the sides **210s** of the upper vessel **210** via a pair of rotating arms **1303** and **1304**. Each end of the arms **1303** and **1304** are configured for independent rotation from the other end to first lift the drum **100** from the lower vessel **220** by rotation of the arms **1303** and **1304** at the coupling to lower vessel **220**. Then, the arms **1303** and **1304** are rotated at the coupling to the base **100b** to invert the drum **100**.

[0090] In FIG. **15A-15C**, the container **200** comprises in a lower portion a drawer **201** that translate away from the lower portion of the container **200** to provide access to material collected therein. In FIG. **15A**, the drawer optionally slides laterally away from the side of the lower portion to provide access for removing material that collected in the drawer after exiting the drum **100** via the perforations **100**. Alternatively, as shown in FIG. **15B**, the drawer **201** may also be a portion of the lower container or an insert therein that is in hinged engagement to swing away from the lower contains and thus act as a shoot to pour the collected material downward toward a conveyor or vessel. A hinged cover **202** may be arranged on an upper portion of the container **200** above the drawer **201** to clean the container **200**. The container **200** may also be configured to be separated for cleaning by a lateral seam that defines an upper and lower vessel, or a vertical seam.

[0091] The drum **100** is removed or inserted in the container **200** laterally via a side portal **200p**

such that it translates in the direction of the principal axis **100a**. Alternatively, the drum **100** can be fixed, and the container **200** can translate laterally. The portal **200p** may be sealed by a cover **116c** that support the shaft **116** coupled to the drum base **100b** for rotating the drum **100**, which then is optionally coupled in rotatory engagement to a motor (not shown) on the opposite side of the cover **116c** from the drum **100**. The opposing side **100e** of the drum **100** may have the cap **117** that is removed to load or unload material. FIG. **15C** illustrates the cover **116c** and drum **100** connected via the shaft **116** with the axis **100a** vertical to fill the drum via end **100e** when the cap **117** is removed. The drum **100** can inverted to be emptied by the same actuator that rotates the cover **116c** after the drum **100** is removed from the container.

[0092] While the invention has been described in connection with several embodiments, it is not intended to limit the scope of the invention to the particular form set forth, but on the contrary, it is intended to cover such alternatives, modifications, and equivalents as may be within the spirit and scope of the invention as defined by the appended claims.

Claims

1. A rotary separation apparatus comprising: a drum having a generally cylindrical shape with a principal axis coincident with the principal axis of a cylinder defined by the generally cylindrical shape, in which the drum has lateral surfaces with a plurality of perforations and a base of the cylinder that is opposed to an upper end of the cylinder; a container that forms an enclosed space around the drum when the drum is disposed with the principal axis in a horizontal orientation; one or more actuators that are operative to modulate the orientation of the principal axis of the cylinder from between an upright and inverted vertical orientation and dispose the drum horizontally in the container; and a means to rotate the drum about the principal axis when the drum is disposed horizontally within the container.
2. The rotary separation apparatus of claim 1 in which the drum has a removable cap that closes an opening in the upper end of the cylinder.
3. The rotary separation apparatus of claim 1 in which the container forms the enclosed space when an upper and lower vessel are brought into mutual engagement at a rim of the upper vessel to a rim of the lower vessel.
4. The rotary separation apparatus of claim 3 wherein the lower vessel is capable of being inverted from an upright horizontal orientation to a downward facing horizontal orientation by the one or more actuators to empty the lower vessel.
5. The rotary separation apparatus of claim 3 wherein the one of more actuators are operative to modulate the orientation of drum from being disposed with the principal axis of the cylinder horizontally within the container to vertically outside the container by rotating the upper vessel away from the lower vessel to dispose both the upper vessel and the lower vessel in an upward facing orientation and then rotating the drum to dispose the principal axis of the cylinder vertically.
6. The rotary separation apparatus of claim 5 wherein the one of more actuators includes a first motor and second motor in which the first motor is operative to rotate the upper vessel away from the lower vessel and the drum is rotated between the vertical and horizontal orientation of the principal axis of the cylinder by the second motor that is coupled to a support frame that is external to a common housing that contains the first motor to rotate the common housing at least 60 degrees.
7. The rotary separation apparatus of claim 6 wherein the means to rotate the drum about the principal axis when the drum is disposed horizontally within the container is a primary motor coupled to the common housing to rotate a shaft that extends through the common housing to connect to the base of the cylinder in which the shaft extends in the direction of the principal axis of the cylinder.
8. The rotary separation apparatus of claim 6 wherein the one of more actuators includes a third

motor that is connected to the support frame and the second motor is mounted at a distal end of a primary arm and the proximal end of the primary arm is connected to the third motor in which the third motor is operative to rotate the primary arm from a horizontal to a vertical orientation before the one or more actuators are operative to modulate the orientation of the principal axis of the cylinder from an upright to an inverted vertical orientation.

9. The rotary separation apparatus of claim 1 wherein the container comprises in a lower portion a drawer disposed to collect matter that exits the drum via the plurality of perforation in which the drawer is configured to translate away from the lower portion to provide access to the collected matter.

10. The rotary separation apparatus of claim 1 wherein the one or more actuators is operative to remove the container laterally via a side portal such that it translates in the direction of the principal axis.
